SALONI DEEPAK

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EDUCATION

University of Notre Dame, Notre Dame, IN, USA
January 2021-present:
2027 (Anticipated) — Ph.D., Physics and Astronomy
2024 — Master of Science, Physics and Astronomy

University of Delhi, New Delhi, India
2020 — Master of Science, Physics (Astrophysics)

St. Xavier's College, Kolkata, India
2018 — Bachelor of Science (Honors), Physics

PUBLICATIONS

- "A global census of metals in the Universe", Deepak, S., Howk, J.C., Lehner, N., Péroux, C. (2025) ApJ, Volume 987, Issue 2, id.199, 24 pp.

ACCEPTED PROPOSALS

- CoI: "Illuminating the Dark Ages of Metal Evolution: An HST Legacy Survey at Cosmic Noon", Hubble Space Telescope (HST), Cycle 32, AR 17862.
- Thesis student: "Metals at Cosmic Noon", Astrophysics Data Analysis Program (ADAP 2024), 24-ADAP24-0169.
- CoI: "A Complete Characterization of Multiphase Circumgalactic Gas in the Cosmic Afternoon through Forward Modeling", HST, Cycle 33, ID 18146.

FELLOWSHIPS, AWARDS AND DISTINCTIONS

2024-2027	Future Investigators in NASA Earth and Space Science and Technology (FINESST) award ($150,000), National Aeronautics and Space Administration (NASA).
2026	Graduate School Professional Development Award : Zahm Professional Development Fund, Graduate School, University of Notre Dame.
2024	American Astronomical Society (AAS) International Travel Grant by AAS (supporting travel to "Baryons Beyond Galactic Boundaries" conference at IUCAA, Pune)
2024, 2026	Graduate Student Government Conference Presentation Grant (CPG), the Graduate Student Government, University of Notre Dame.

2024	Graduate School Professional Development Award : Notebaert Professional Development Fund, Graduate School, University of Notre Dame.
2018	Ashish Palit Memorial Prize for highest graduating marks in B.Sc. Physics, Department of Physics, St. Xavier's College, Kolkata.
2016-2017	Research scholarship at National Initiative on Undergraduate Science (NIUS), Homi Bhabha Centre for Science and Research, Tata Institute of Fundamental Research.
2015-2020	INSPIRE Scholar: 5-year scholarship awarded by the Department of Science and Technology, Ministry of Science Technology, Government of India.

CONFERENCE TALKS, POSTERS

2026	Invited talk - "Bridging the Gap at Cosmic Noon - Metals in the CGM" at the workshop "Galactic Metal Cycles", University of Notre Dame, Kylemore, Ireland.
2026	Contributed talk - "BRIDGE I - Metals in the CGM at Cosmic Noon" at the conference - "Celebrating 40 years of Damped Lyman alpha Absorbers", Kavli IPMU, University of Tokyo, Japan.
2025	Science talk, 31st Annual Keck Science Meeting - "Towards a Global Metal Budget of the Universe - the missing link at Cosmic Noon", University of California, Los Angeles.
2025	Astrophysics Seminar - "Towards a Global Metal Budget of the Universe - the missing link at Cosmic Noon", Department of Physics and Astronomy, University of Notre Dame.
2024	Contributed talk - "A global census of metals in the Universe" at the conference - "Baryons Beyond Galactic Boundaries", IUCAA, Pune, India.
2024	Contributed talk - 'The global distribution of metals in the Universe' at the Figuring out Gas and Galaxies in Enzo (FOGGIE) retreat, Michigan State University.
2023	Poster presentation: 'A global census of metals in the Universe' at the conference - "Oases in the Cosmic Desert: Understanding the Structure of the Circumgalactic Medium", School of Earth and Space Exploration at Arizona State University.
2019	Project presentation - 'Determining the host luminosity of a gravitationally lensed AGN', IUCAA, Pune, India.

WORKSHOPS, SCHOOLS

2025	AstroAI workshop, Center for Astrophysics, Harvard & Smithsonian (CfA).
2024	Code/Astro software development workshop, Northwestern University. Development and packaging of absorption line finder tool pyALF (https://github.com/sameeresque/pyALF.git).
2023	Michigan Cosmology Summer School 2023, University of Michigan, Ann Arbor.

2023	Jupyter notebook workshop: Developing accessibility tools in Astronomy on the "Day of Accessibility", Space Telescope Science Institute (STScI).
2022	Summer School in Statistics for Astronomers XVII (2022), Penn State University - Eberly College of Science.
2019	Visitor Student Research Programme, The Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India.
2019	Visitors Programme - "Recent Advances in Physics", Department of Physics and Astrophysics, University of Delhi.
2018	Star-DBT workshop on Selected Topics on Advances in Physics-Biology Interface, the Department of Biotechnology, Ministry of Science Technology, Government of India.

OTHER RESEARCH EXPERIENCE

2020-2021	**Polytropic gas atmosphere around a spherically symmetric gravitating core** Research Advisor: Prof. T. R. Seshadri, University of Delhi, New Delhi, India
2019-2020	**Magnetic field of the Milky Way from Faraday rotation of pulsar signals (M.Sc. Thesis)** Thesis supervisor: Prof. T. R. Seshadri, University of Delhi, New Delhi, India
2019	**Determining the host luminosity of a gravitationally lensed AGN** Research Advisor: Dr. Anupreeta More, The Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India
2018	**The Dynamics of Spherical Bondi Accretion: An Overview (B.Sc. Thesis)** Thesis supervisor: Prof. Suparna RoyChowdhury, Department of Physics, St. Xavier's College, Kolkata, India
2016-2017	**Estimation of extent of convective cores in pulsating stars** Research Advisor: Prof. Anwesh Mazumdar, Tata Institute of Fundamental Research, India

POSITIONS HELD

Spring 2021	Teaching Assistant : Intermediate Mechanics, Department of Physics and Astronomy, University of Notre Dame.
Fall 2021—present	Research Assistant : Department of Physics and Astronomy, University of Notre Dame.

SERVICES

2024-2026	Mentor under the mentorship program for undergraduate students at the Department of Physics and Astronomy, University of Notre Dame. Mentee: **Anandini Mitra**.

2024	Judge at the Northern Indiana Regional Science and Engineering Fair (NIRSEF).
2023-2024	Chair of International Committee, The Graduate Physics and Astronomy Society, Notre Dame.
2023	Mentor under the mentorship program for graduate students in the Department of Physics and Astronomy at the University of Notre Dame.
2016-2017	Mentor under the National Service Scheme for primary school students, West Bengal, India
2015-2018	Member and volunteer for the National Service Scheme, the Ministry of Youth Affairs and Sports, Government of India.

TECHNICAL SKILLS

Programming Languages	Python, Java, C++, FORTRAN.
Softwares	MESA (stellar modeling), glafic (gravitational lens modeling), SAOImage DS9, Cloudy (spectral analysis and ionization modeling), Pypeit (spectra calibration and reduction).
Data and Code Management	GitHub (https://github.com/sdeepak1997).

LANGUAGES

English (native speaker, reading, writing).
Hindi (native speaker, reading, writing).
Bengali (reading, writing, speaking).

REFERENCES

J. Christopher Howk Professor Department of Physics and Astronomy University of Notre Dame +1 (574) 631-8594 Email: jhowk@nd.edu	Nicolas Lehner Research Professor Department of Physics and Astronomy University of Notre Dame +1 (574) 631-5755 Email: nlehner@nd.edu